MECHENG 706
Mechatronics Design Projects

Course & Projects Information

Lecture 1, Week 1

THE UNIVERSITY OF AUCKLAND
Te Whare Wānanga o Tāmaki Makaurau
NEW ZEALAND

ENGINEERING

MECHENG 706
Mechatronics Design Projects

Lecturers:

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THE UNIVERSITY OF AUCKLAND
Te Whare Wānanga o Tāmaki Makaurau
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ENGINEERING

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ENGINEERING

MECHENG 706 Course Assessment

Project 1, 32.5%, in group of 4, due Friday (24 Apr), Week 6

Farm Tilling Service Proof-of-Concept (involves localisation and mapping)

- Path Following (20%)
- Time taken (20%)
- Mapping and Coordinates (20%)
- Summary of Results (20%) and Contract Proposal (20%)

(JD is in charge of Project 1)

Project 2, 32.5%, in group of 4, due Friday (5 Jun), Week 12

Behaviour Control Robotics (for firefighting and obstacle avoidance)

- Fire detection & extinguishing (20%)
- Time taken (20%)
- Obstacle avoidance (20%)
- Report (40%), where 30% individually assessed & 10% in group

(PX is in charge of Project 2)

Weekly Interview, 5%

- Attendance (50%)
- Discussion (50%)

(Each group is interviewed once a week for 10 minutes during the 2nd lab session)

Peer evaluation as a weighting for all the group marks

- **The form available at Canvas/Files/Course & Project Info**

Exam, 30%

(you must sit to pass the course, otherwise, will get a DNC)
(Open-book, 2 hours, in-person, invigilated)

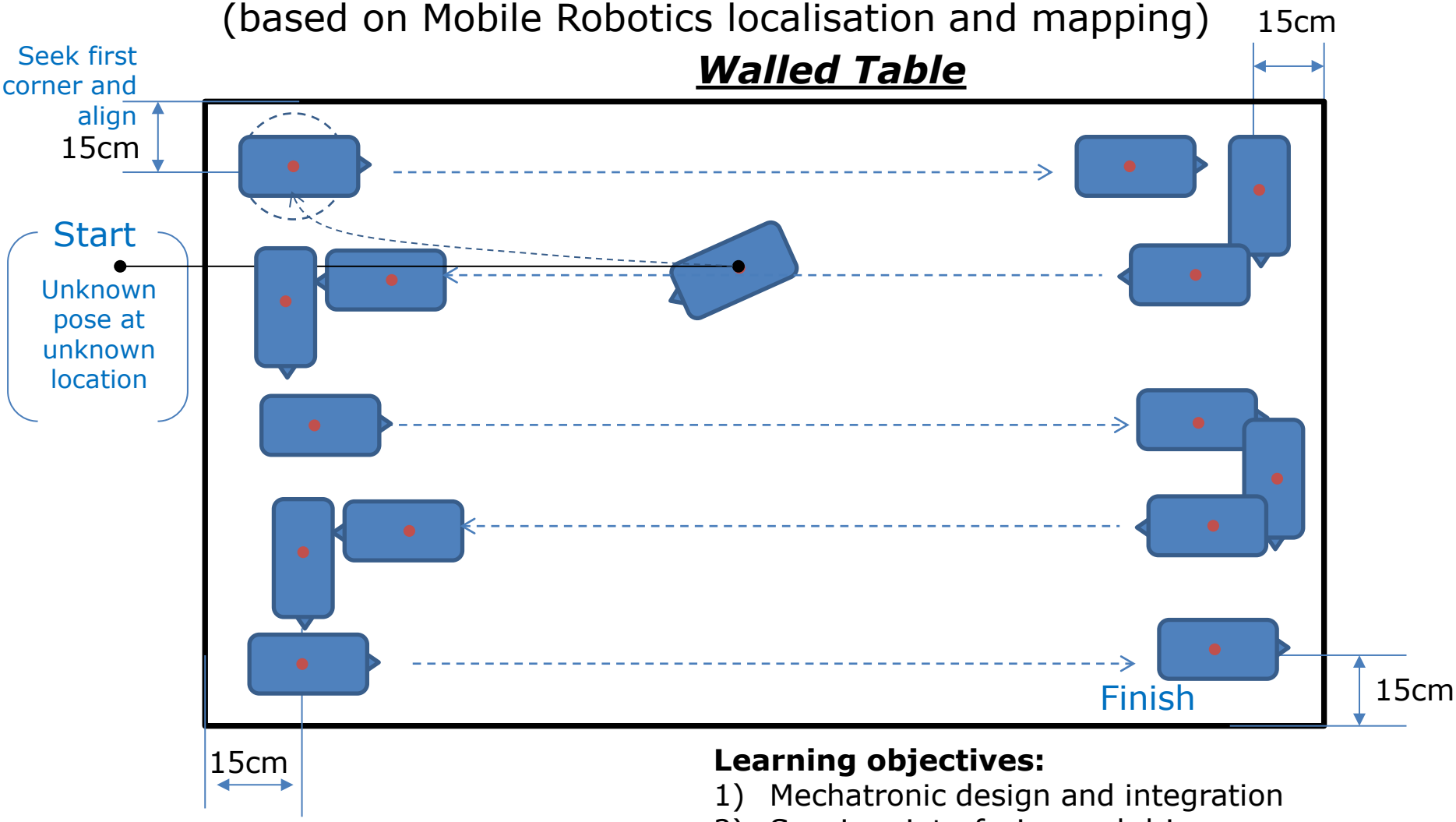
Project 1: Real world inspiration

- Your Company: AgriService Ltd. provides contract and rental services to farm landowners.
- Your Team: In charge of providing tilling contract service
- Your Tasks:
  1. Develop new innovative automated solution for farm tilling, by testing out proof-of-concept technology in a lab environment. Summarize results.
  2. Draft a proposal with budget for providing contract service.

- Assumptions and Constraints
  - Agricultural plots are mainly rectangular.
  - Restricted to the resources provided in your MECHENG 706 kit.

Project 1 Farm Tilling Service Proof-of-Concept

(based on Mobile Robotics localisation and mapping)



Learning objectives:

1) Mechatronic design and integration
2) Sensing, interfacing and driver
3) Motor, interfacing and driver
4) Closed-loop control implementation
5) Real-time programming
6) Mobility mechanism
7) Application of learned professional skills

Assessment Criteria:

1) Path Following, 20%
2) Time taken, 20%
3) Mapping and Coordinates, 20%
4) Report, 40%

Rubrics – Project 1

Path following± Deviation (20%)
± 10 mm or less, 10 points
± 12 to 14 mm, 9 points
± 14 to 16 mm, 8 points
± 16 to 18 mm, 7 points
± 18 to 20 mm, 6 points
± 20 mm or larger, 5 points
Base point, 4 points
*(Yaw angular deviation> ±**8**^{o}, penalised)*

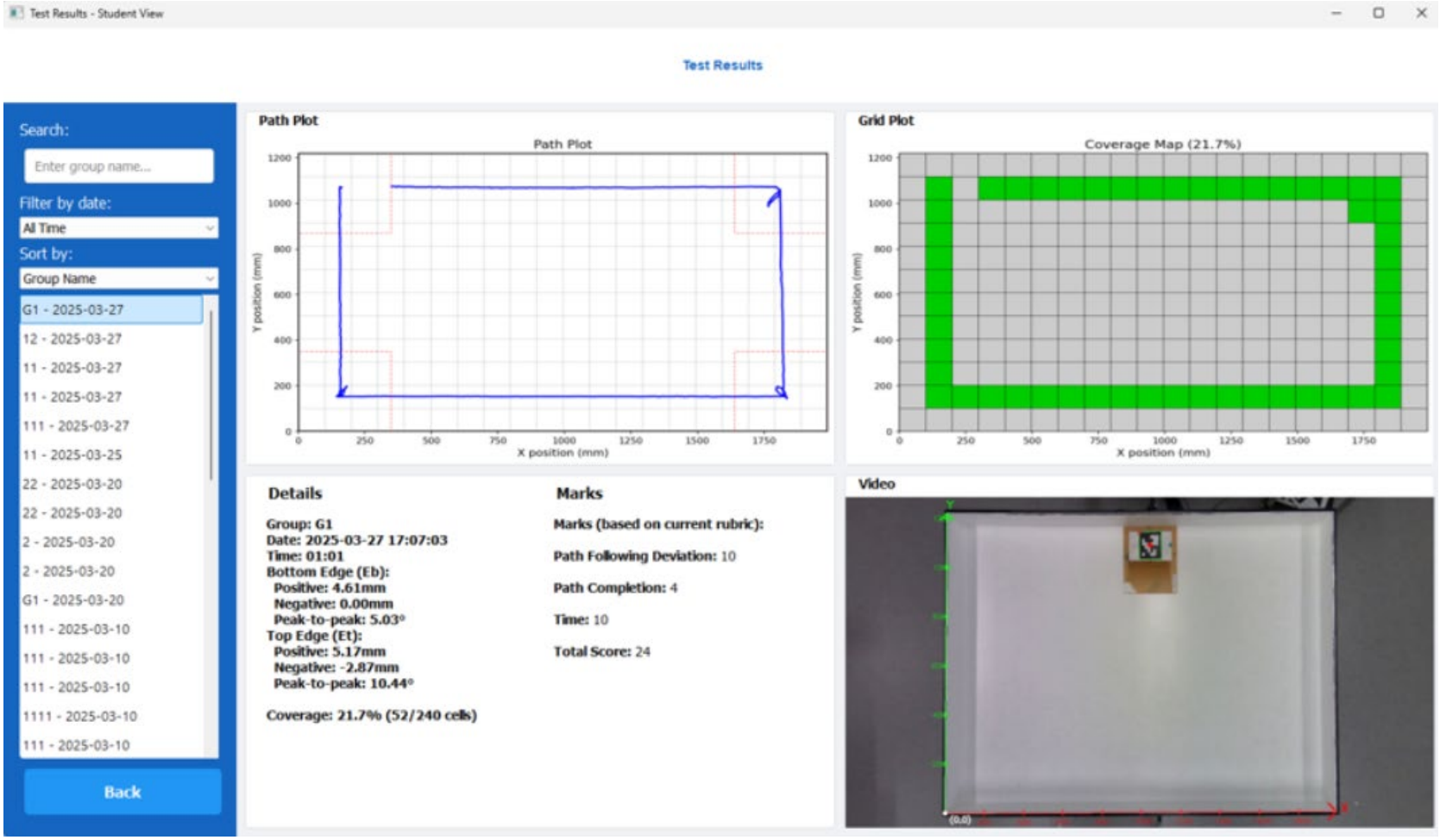
Time (20%)
90 sec or less, 10 points
90 to 100 sec, 9 points
100 to 110 sec, 8 points
110 to 120 sec, 7 points
120 to 130 sec, 6 points
130 sec or above, 5 points
Base point, 4 points

Path Completion (10%)
100% complete, 10 points
75% to 100% complete, 8 points
50% to 75% complete, 6 points
Base point, 4 points

Coordinates (10%)
100% Generated map, 10 points
75% to 100% Generated map, 8 points
50% to 75% Generated map, 6 points
Base point, 4 points

Note: Base points are awarded only to those whose demonstration shows evidence of effort, but fails to achieve objectives.

Assessment software



Assessment software captures video stream from an overhead camera to find

- Errors from straight line
- Area covered
- Yaw deflection (in the distance and angle)
- Time lapsed between start and end of the run

Summary of Results, 20% of the project

Each student presents 1 page summary of any aspect of the project work

- Integration of sensors/ actuators
- Path planning and Mapping
- Control System
- Software architecture and hardware considerations

Present plots, figures and brief explanation.

All figures, results plots and graphics must be original work made in context of your project.

Only text description for any section will be penalised for lacking graphical illustrations.

Marked individually, but a group penalty will be applied if

1. There is a technical error in project execution (e.g. wrong control logic, or wrong calibration method).
2. Same information is repeated in different sections.
3. If format, figure numbering, equation numbering etc are inconsistent.

Submit as a single PDF file for the group.

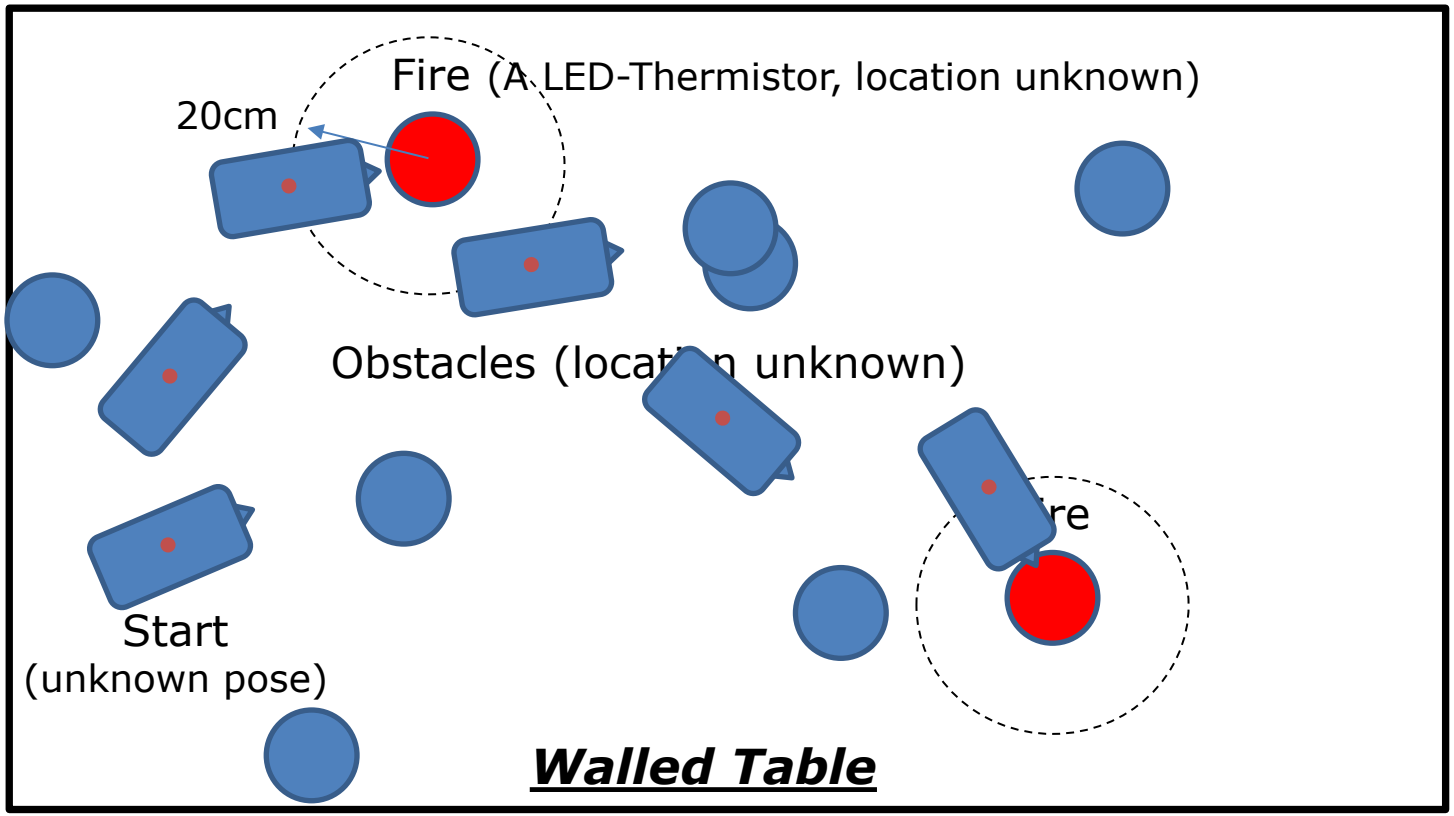
Proposal for Tilling Service (20 % of the project)

- Submit a contract proposal to a farmland owner for tilling a 1000m × 500m section of land. NOTE: YOU SHOULD DRAFT THIS ASSUMING YOUR PROPOSAL IS READ BY YOUR CLIENT
- Should be 4 pages as follows:
- 1st page:
  - Statement of work
  - Responsibilities
  - Safety consideration undertaken
- 2nd page: Quote (with a breakdown including but not limited to equipment use/ hire, consumables, labor). It is expected that the total cost should not exceed $15 K.
- 3rd page: A table listing foreseeable hazards (for safety, equipment, legal, etc.) and its mitigation.
- *4th page: List the references used. List out any websites, or other material (e.g. equipment costs) you referred to develop the proposal/ quote. This page is not part of proposal but to show that you developed that proposal through due diligence.*

Marked as a group

Project 2 Behaviour Control Robotics

(for autonomous navigation, firefighting and obstacle avoidance)



Assessment Criteria:

1) Fire detecting/extinguishing, 20%
2) Time taken, 20%
3) Obstacle avoidance, 20%
4) Report, 40%

Learning objectives:

1) Mechatronic design and integration
2) Sensing, interfacing and driver
3) Motor, interfacing and driver
4) Real-time programming
5) Behaviour based control
6) Fuzzy logic control
7) **Agile project management**

The Challenge:

The robot you design/build, starting from an unknown location/orientation, will navigate by itself. It will need to find two fires and extinguish them as quickly as possible while avoiding any collision with 2 fires, an unknown number of obstacles and 4 walls. Most of obstacles are static but one or two may move on the table. The robot must stop moving by itself once the 2nd fire has been put out.

The Constraints:

The fires and obstacles are cylindrical objects in diameter of 10 cm and height of about 12 cm. They are randomly placed on the table.

A fire (i.e., a LED-Thermistor combination), sitting atop the cylinder, can be put out by a fan that needs to be turned on for 10 seconds before the thermistor triggers to switch off the LED. Should the robot manage to extinguish the fire in less than 10 sec, it can move away and forward immediately.

A fire allows to be extinguished only when the robot moves into the vicinity of the fire, which is defined by a distance of 20 cm from the center of the fire to the center of the robot.

Rubrics – Project 2

Fire detection/extinguish (20%)
2 fires extinguished, 10 points
1 fire extinguished, 1 detected, 9 points
1 fire extinguished, 8 points
1 fire detected, not extinguished, 7 points
0 fire detected, 6 points
Base point, 4 points

Obstacle avoidance (20%)
No collision, 10 points
1 collision, 9 points
2 collisions, 8 points
3 collisions, 7 points
4 collisions, 6 points
5 collisions, 5 points
Base point, 4 points

Time (20%)
60 sec or less, 10 points
60 to 70 sec, 9 points
70 to 80 sec, 8 points
80 to 90 sec, 7 points
90 to 100 sec, 6 points
100 sec or above, 5 points
Base point, 4 points

Report (40%)
Technical contents, 70%
(eg method, math, symbols, description,…)
Presentation 15%
(eg figure, pseudo code, flowchart, diagram,…)
Formatting/Language, 15%
(eg grammar, font, logic, pagination, …)

Note:
(1) Base points are awarded only to those whose demonstration shows evidence of effort, but fails to achieve objectives.
(2) The key design challenge is fire detection/extinguishing. When the demo is marked, Time will be given Base Points if the robot fails completely in firefighting and obstacle avoidance.

Project Report 2, 40% of the project

Suggested Table of Contents (in a single report in PDF)

- Fire sensing/extinguishing system (sensor/motor/circuit/ programming etc), 30%, by student 1
- Obstacle sensing system (sensor/circuit/programming etc), 30%, by student 2
- Behaviour control (BBC/FLC/FSM, codes etc), 30%, by student 3
- System integration (system configuration in SW and HW, MCU pin allocations and functions, flow charts, architecture, pseudo codes etc), 30%, by student 4
- **Project management, 10% in group**

For each subsystem, write about "what", "how" and "results".

Each chapter including Project Management is restricted to **two pages,** single line spacing and 11 font size of Arial. *Introduction, Conclusion and* ***Appendix are not required and not marked if given.***

"What" – what (sub)system you have worked on, including problem statement, scenarios, requirements, assumptions etc.
"How" – how the (sub)system was designed, interfaced to MCU, driven by program, tested/calibrated, etc. How the entire system was integrated.
"Results" – evidence of how the (sub)system works, including data, charts, diagrams, discussion of results etc.

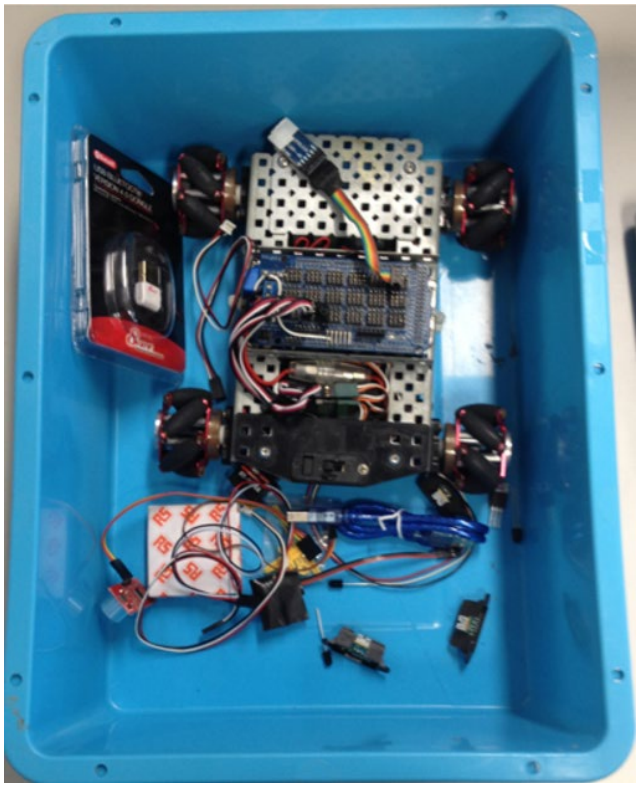
Project Management and Marking Rubrics

- 2 page only, in group
- Have you applied any project management skills/knowledge learnt in *ENGGEN 303 Managing Projects and Innovation* into this project? If yes, what is it and how? If not, summarise the learnt skills/knowledge that may be applied to this project. (25%)
- Reflect how you have practised ***Agile*** product development technique (Lecture 2/Week 1) into this project. (25%)
- Describe any software (e.g., Jira, Microsoft Projects) used, or evaluated for potential use in this project. (25%)
- State who has contributed to which part of the project, and to what extent. (25%)

Write this chapter as specific to the project as possible, with the support of evidence such as charts, diagrams, plots etc.

List of Parts

- 1x Mobile base (incl. Arduino Mega, shield, battery, 4 motors)
- 1x Small Servo
- 4x 2D120X or 2Y0A41SK (IR Sensor, medium range)
  - Range 40-300mm
  - with 2 cables
- 4x 2Y0A21 (IR Sensor, long range)
  - Range 100-800mm
  - with 2 cables
- 1x BNO08X motion sensor
  (use single axis gyro only)
- 1x HC-SR04 (Ultrasonic Sensor)
- 4x Phototransistors
- 1x Fan
- 1x USB Cable
- 1x Wireless Module
  (incl 2xHC12, 1xYP-05, 1 USB A to USB mini cable)
- 1x MOSFET (FQP30N06)
- 1x 5V regulator (7805 5V 1A TO-220)
- 1 x Lipo battery (with a smart charger)
- (+ some brackets etc available in lab)
- (+ self-made 3D printed brackets are allowed)



Note: Only maximum 2 IR sensors of each type may be used in the final design solution for both projects.

Functions of Major Parts

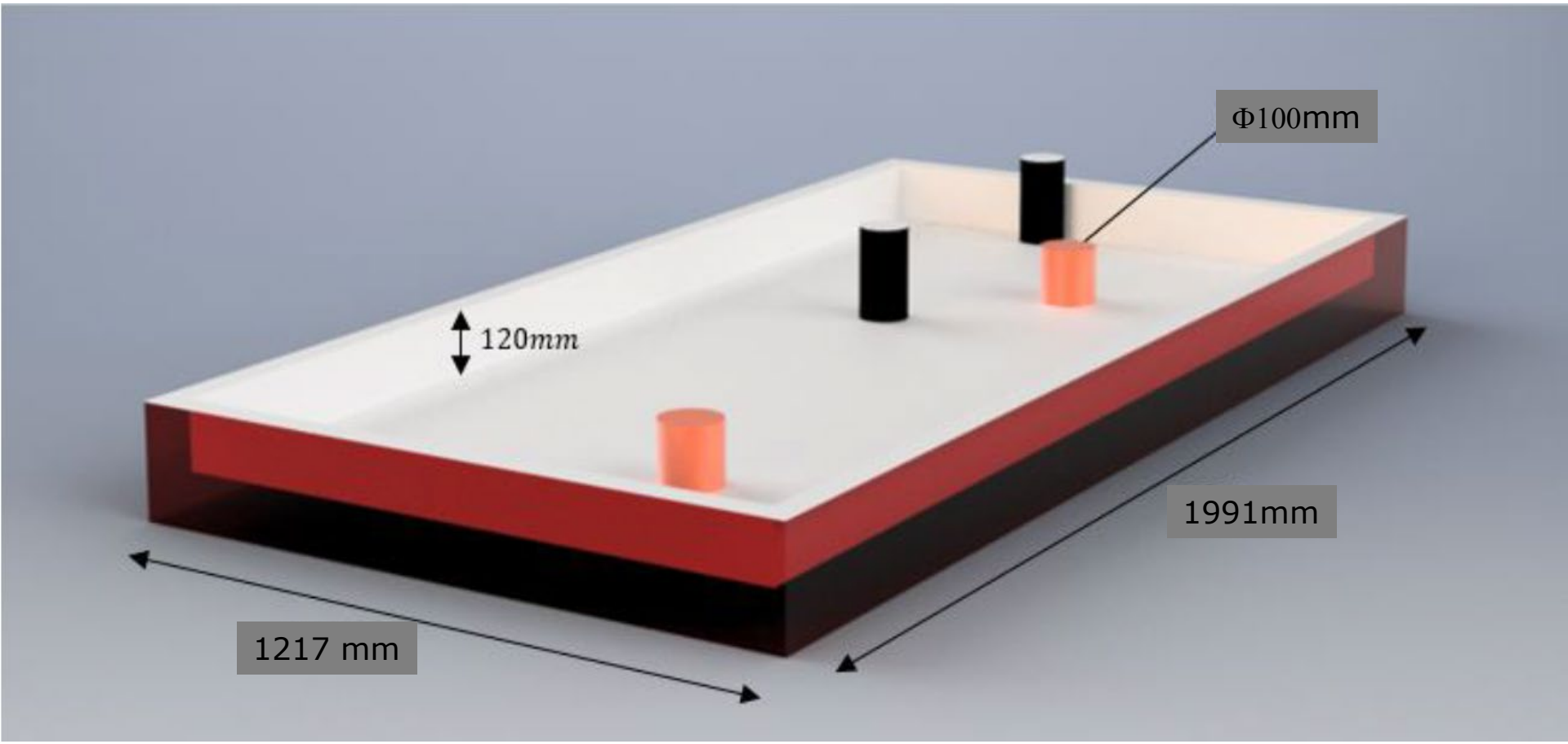
1. One Mecanum wheeled mobile base (incl 4 motors without encoders, 1 Arduino board, 4 motor drives)
   (for omni-directional movement. Don't modify the chassis and the parts on it.)
2. Use only one axis gyroscope of BNO08X ***(for angle and angular velocity)***
3. Two long range IR sensors and Two medium range IR sensors ***(for object and wall detection)***
4. One sonar sensor ***(for the purpose of detecting the walls and objects)***
5. One servo motor (***used to rotate sensor(s) or the fan)***
6. Four phototransistors ***(for fire detection)*** and 1 small electric fan ***(for fire extinguishing)***
7. One wireless module ***(used for logging robot motion data back to PC/Laptop)***
8. Other discrete electric components are available in lab, e.g. resistors, nuts, bolts, brackets for various minor functionalities.

Notes:

1. It is not obligatory to use all the parts supplied for the purpose of project requirements.
2. You are not allowed to use any major parts other than supplied.
3. Minor 3D printing parts are allowed.
4. Sample codes **(*for sets of basic robot movements, Sensors, Behaviour Control etc*)** can be found at Canvas, in the folder **"Files/Sample Codes"**

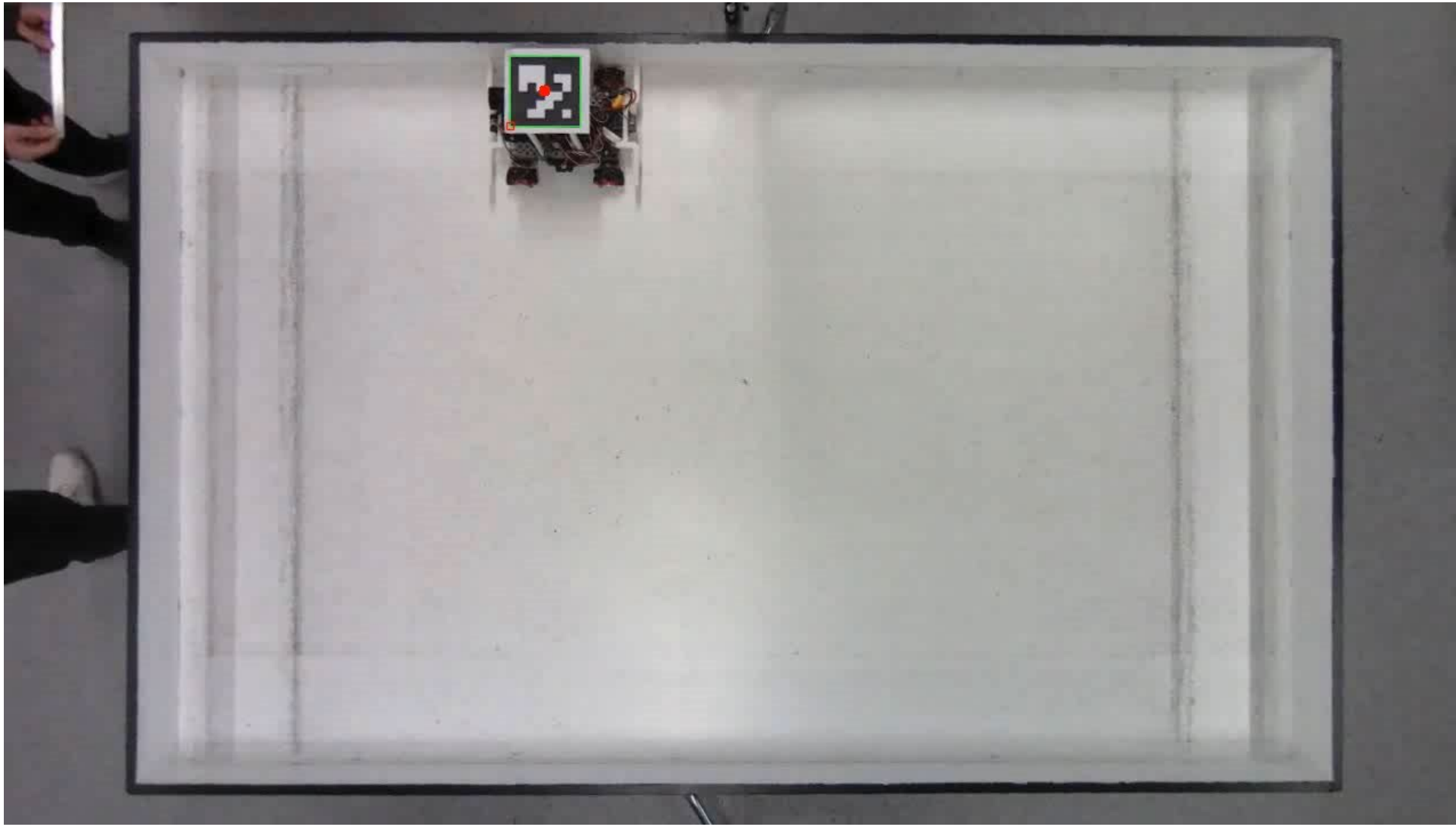
Arena, Objects and Fires

(The table is placed in 405.564 at all the times.)

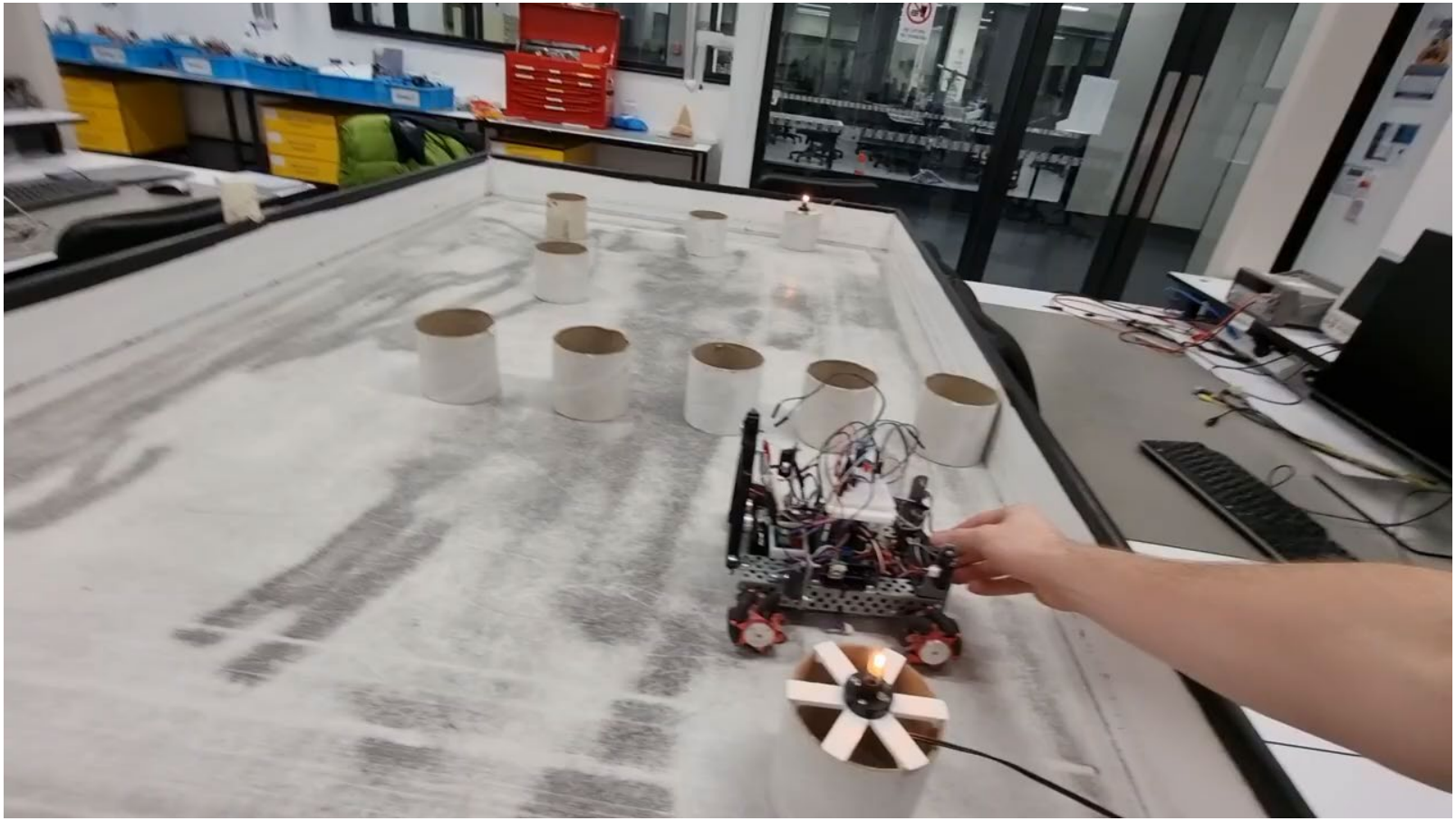


A Sample Robot
(SLAM, Area Covering, Obstacle Avoidance and Firefighting)

A Sample Demo (Project 1, 2024)



A 2024 Sample



https://youtu.be/AcPfPGBH5HM

Tutorials on Sample Codes of Robot Motion, Sensors & Wireless Module

1. Hardware connection, programming and basic functionalities (How Arduino and Shield are connected? Key input/output pins and their functions. How is Arduino programmed? Arduino programming environment.)
2. Motors interface, drive and Arduino programming of Mecanum robot (What are the motors/specs? How 4 motors are interfaced to Arduino and how they are commanded by Arduino? Codes for the robot moving forward and turning on spot.) - a simple basic design
3. IR range sensor interface, and Arduino programming for measurement of object distance (What is the sensor/specs? What does it do? How is the sensor connected to Arduino, what simple codes are needed to get in the signals, how are they processed?) – a simple basic design
4. BNO08X motion sensor (Use single gyroscope only), interface, and Arduino programming for measurement of the angular velocity (What is the sensor/specs? What does it do? How is the sensor connected to Arduino, what simple codes are needed to get in the signals, how are they processed?) – a simple basic design
5. Fire sensor interface and Arduino programming for detection of a fire (What is the sensor/circuit? What does it do? How is the sensor connected to Arduino, what simple codes are needed to get in the signals, how are they processed?) – a simple basic design
6. Wireless Module for logging robot states in Arduino back to PC (including the pin connection and sample codes)

Tutorial materials available in Canvas folder, Files/Laboratory

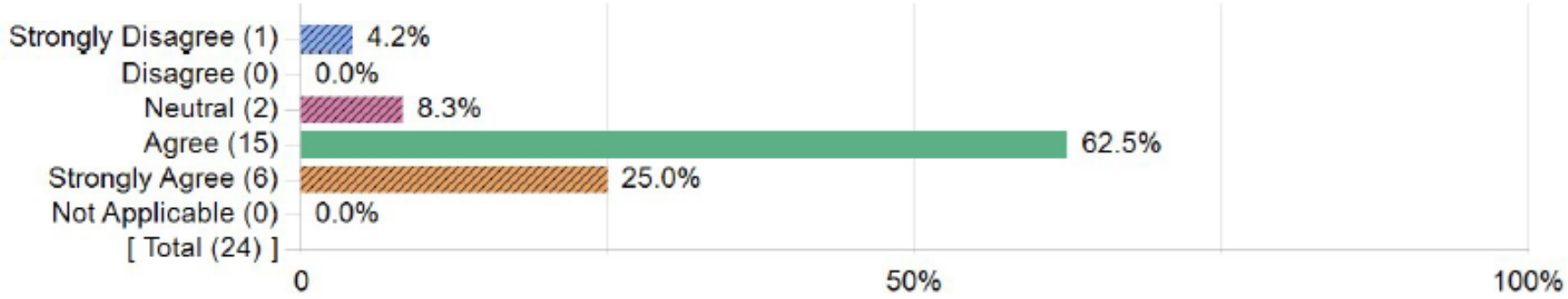
Week (Starting on Mon date)	Lecture 1 3-4pm, Mon, 106-204	Lecture 2 4-5pm, Thu, 421W-201
1 (2 March)	Course/projects description (PX, JD)	Mechatronics design (JD) (agile methodology)
2 (9 Mar)	Mecanum robotics (JD) (mobility principle, kinematics, example)	Mecanum robotics (JD) (closed-loop control, line following example)
3 (16 Mar)	Applied Kalman filters principle (JD) (1D problem principle)	Applied Kalman filters example (JD) (1D problem example, implementation)
4 (23 Mar)	Advanced Kalman filter/SLAM (JD) (sensor fusion, SLAM)	*Project 1 Assessing software (JD/HH)*
5 (30 Mar)	Advanced Kalman filter/SLAM (JD) (sensor fusion, SLAM)	Homogenous transformation matrix (JD)
Mid-semester Break		
6 (20 Apr)	Homogenous transformation matrix (JD)	**Project 1 Demo (JD)**
7 (27 Apr)	ANZAC Day	Mechatronics design (PX) (case studies)
8 (4 May)	Behaviour-based robotics (PX) (single behaviours)	Behaviour-based robotics (PX) (networked behaviour, case study, coding)
9 (11 May)	Fuzzy logic control (PX) (fuzzy sets, membership functions)	Fuzzy logic control (PX) (fuzzy rules, fuzzy inference)
10 (18 May)	Fuzzy logic control (PX) (design, case study, coding; MATLAB toolbox)	AI for mechatronics (PX) (artificial neural networks – architecture)
11 (25 May)	AI for mechatronics (PX) (artificial neural networks – Training)	AI for mechatronics (PX) (artificial neural networks – examples)
12 (2 Jun)	AI for mechatronics (PX) (transformer architecture)	**Project 2 Demo (PX)**

SET Course Report 2025

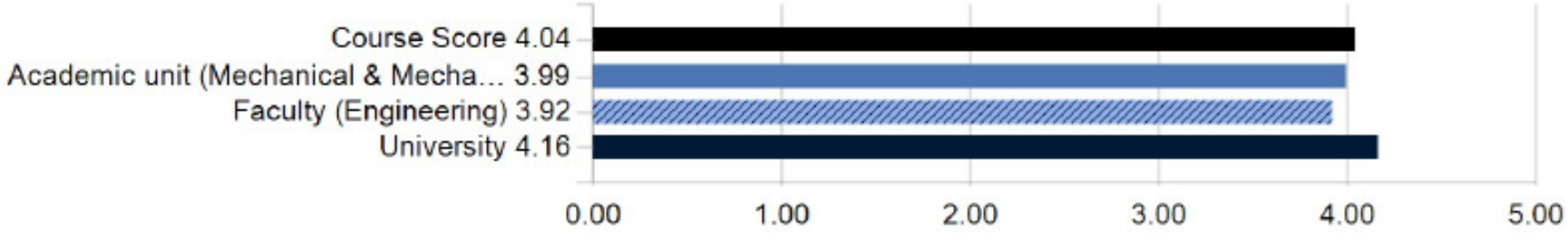
Course Overall Question

	Resp	%GA	%Neutral	%GD	Mean	Median	SD	SIR
Overall, I was satisfied with the quality of this course.	24	87.5%	8.3%	4.2%	4.04	4.10	0.86	0.25

Overall, I was satisfied with the quality of this course.



Overall, I was satisfied with the quality of this course.



SET Feedback 2025

Most Negative Feedback

"All the hardware issues haunted me in my sleep several times."

"The number of times hardware failed in the projects was challenging; it cost us a lot of time and effort."

"I felt that the Kalman filter content was quite hard to grasp. It would be good to have some sort of practice or workshop/lab where we try implement our own Kalman filter?"

"Overall, the lecture content and what we actually do on the robots is extremely disconnected. We have used SOME of the lecture content on our robot, but that was more of the simple ideas such as Kalman filters."

SET Feedback 2025

Most Positive Feedback

"The projects were fun and a good way to combine a lot of the things we learned throughout our degree. General lecture topics are interesting."

"Hands–on projects and lecture content that was relevant to the projects were super helpful. also, having labs twice a week and getting to ask questions to the tutors was helpful."

"Overall, the teaching team was great, taught helpful things, and I appreciated the changes they made based on previous years to make our year have a better experience."

"Projects. Hands–on applied mechatronics projects are super helpful for consolidating what we have learned thus far. It was nice to learn things like behavioural control and kalman filter and be able to easily implement them on our robot to better understand how they work in practice."

"The projects were really great, and a lot more interesting than the other design courses up to this point. I enjoyed the continuation between the two projects with using the robot."

Continuous Improvements Over Years

- Two single projects (32.5%) ➔ 1 big project (70%).
- Workloads reduced due to 2 smaller independent projects.
- Design challenges balanced between 2 projects.
- Tutorials and worked samples provided for basic robotic motion and sensing.
- Project website requirement removed.
- Project report shorter (2 pages each student, per project).
- Individual assessment increased (>30%).
- Marking rubrics given to students in the beginning.
- Lecture contents tailored/streamlined more for projects.
- AI techniques introduced for advanced mechatronics.
- Final exam ➔ Test (during Week 11).
- Lecturers responsible for all project marking.
- Lecturers attending lab sessions & meeting with students weekly.
- "Mechatronics Design Projects" ➔ "Mechatronics Design" (2023)
- Professional skills & project management techniques in the context of Mechatronics Design (2023).
- More spare sensors and other key parts stocked (2025)
- Three testing platforms made (2025)
- Advanced assessing software developed (2025)
- A new motion sensor (BNO08X) to replace old obsolete single axis gyro (2026)

Get it Started Now

- Sign up for your group of 4 students on CANVAS by the end of today (5pm, Wednesday, 4 March).

Stream 1 **TA: Hesam**	**Stream 2** **TA: Tamara**	**Stream 3** **TA: Siqi**	**Stream 4** **TA: Yitong**
9-11am, Mon & **9-11am, Thu** **Lab 405.552**	**9-11am, Mon &** **9-11am, Thu** **Lab 405.564**	**9-11am, Wed &** **9-11am, Fri** **Lab 405.552**	**9-11am, Wed &** **9-11am, Fri** **Lab 405.564**
Group 1 – 6, 25	Group 7 - 12	Group 13 - 18	Group 19 – 24

- Drop-off sessions (unsupervised):

  4-6pm, Mon, 405.552, starting from Week 2
- Week 1, you will need to attend the 2nd lab session only.
- 3D Printing Laboratory:

  4-6pm, Wed, 405.347, starting from Week 2, except 1 April

  "How to use 3D printer", available at Canvas/Files/Laboratory
- Laboratory 405.564 and 405.552, Lab technician: Terry Huang
- After-hour lab access: Refer to the 1st Canvas announcement.

MDLS Health & Safety

Online Induction:

- Must be completed before 23:59pm, Wednesday, 4 March, Week 1
- MDLS staff invite all students to complete the MDLS CANVAS Induction (FoE MDLS Induction).
- Students must accept the invitation sent to their university email address, complete all required modules, and achieve a perfect score in the quiz.
- The required quiz score is 10 out of 10.
- Students must complete (1) the general induction and (2) specific room induction, i.e., 405.552 or 405.564.

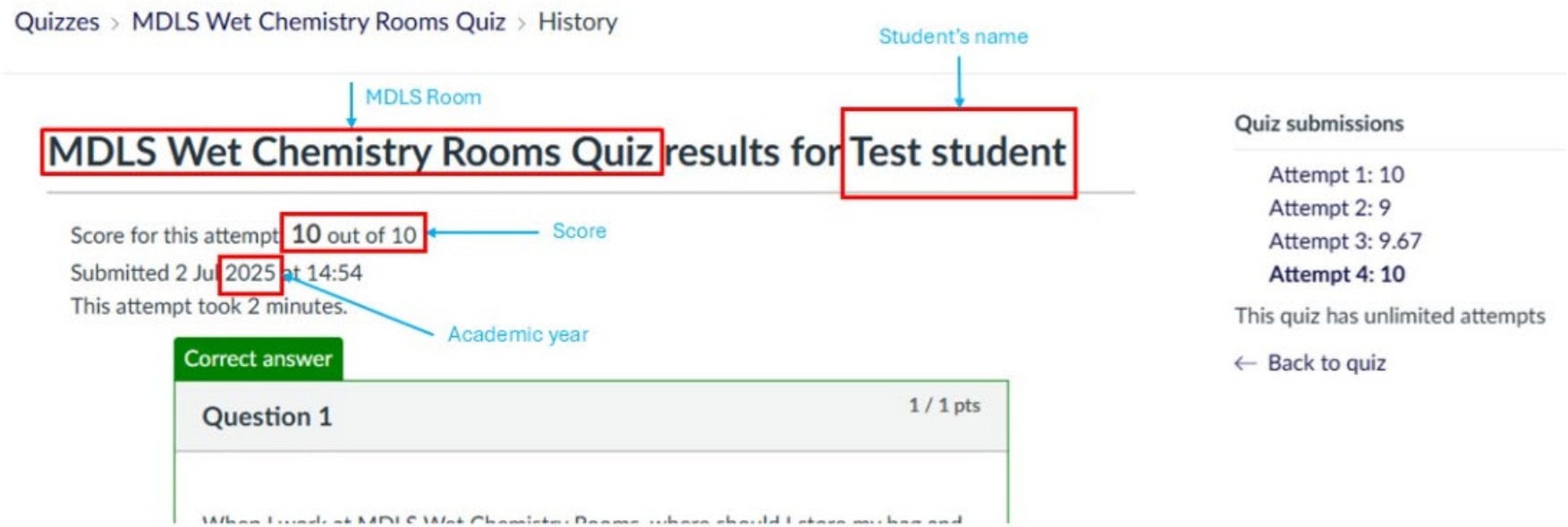
Verification, via Canvas assignment titled "*MDLS Canvas Induction Proof*"

- Upon completing the induction, take a screenshot of the quiz results
- Upload the screenshot, including your name, MDLS room (name of the quiz), score (10 out of 10) and year of completion.

In-person Induction (you must attend the in-person session below):

- Stream 1 & 2, 9-11am, Thu, Week 1, Room 405.452 & 405.564
- Stream 3 & 4, 9-11am, Fri, Week 1, Room 405.452 & 405.564

Example of screenshot



Check the following submission details:

1. Student's name
2. MDLS room – name of the quiz
3. Score must be 10 out of 10
4. Academic Year – MDLS induction is valid only for the current academic year. Regardless of the month it was completed, verification should confirm that the quiz submission date falls within the current year.

A TWO-LANE APPROACH TO MANAGE USE OF AI TOOLS IN ASSESSMENT (2026 New)

Type of Assignments	Lane 1: Controlled assessment of learning (These may or may not involve AI, depending on the discipline and task .)	Lane 2: Uncontrolled assessment for learning (Students may use generative AI to assist in the creation of an artifact, where that artifact is assessed.)
Project 1		
Peer Evaluation		
Results Summary		Yes
Contract Proposal		Yes
Demonstration	Yes	
Project 2		
Peer Evaluation		
Report		Yes
Demonstration	Yes	
Weekly Interview		
Exam	Yes	

References:

- UoA Two-lane approach to assessment
- Advice for students on using Generative Artificial Intelligence in coursework

MME Department's statement on use of AI Tools

Generative AI, like any other tool, is a technology that can support your learning and enhance your work, but it must never substitute for your own critical thinking; you are expected to understand underlying concepts well enough to cross-check AI results and defend the validity of your engineering decisions.

Here, and in your career, you must uphold standards of Professional Accountability:

- **Final Verification Duty:** Manually verify all AI outputs for technical accuracy and relevance.
- **Legal Ownership:** Accept full legal responsibility for designs, regardless of the tools used.
- **Data Stewardship:** Protect sensitive information by avoiding data uploads to public AI models.
- **Ethical Integrity:** Honestly disclose AI usage, and base reports on verified sources of data and information.

The End

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ENGINEERING